

PROJECT REPORT



Techhansa Retail

IT Hardware Bulk & Franchise CMS Platform

1. Executive Summary

The proposed system is a centralized web-based platform designed to streamline IT hardware distribution across two distinct business models: Direct B2B Corporate Sales and Franchise Store Operations. By utilizing a Role-Based Access Control (RBAC) architecture, the platform provides dedicated dashboards for corporate buyers, verified franchise owners, and system administrators, ensuring complete visibility over inventory, bulk orders, invoicing, and logistics.

Core Objectives

- Create a trusted, brand-focused CMS website to showcase bulk IT hardware solutions to corporates, institutions, and resellers.
- Include a dedicated, compelling section designed to attract and convert visiting entrepreneurs into franchise partners.
- Automatically display customized content, pricing, and features across three distinct user roles without manual intervention.
- Provide a single control panel to easily manage franchise approvals, user permissions, site content, and analytics.

Dialogue line

Your Growth. Our Hardware

2. Landing Page Architecture

The homepage is structured as a single top-to-bottom scroll journey. Each section has a specific job: pull the visitor in, build brand trust, showcase inventory, justify bulk purchasing, and then pitch the franchise opportunity before closing with support information. The diagram below shows this flow exactly as it appears on the page.

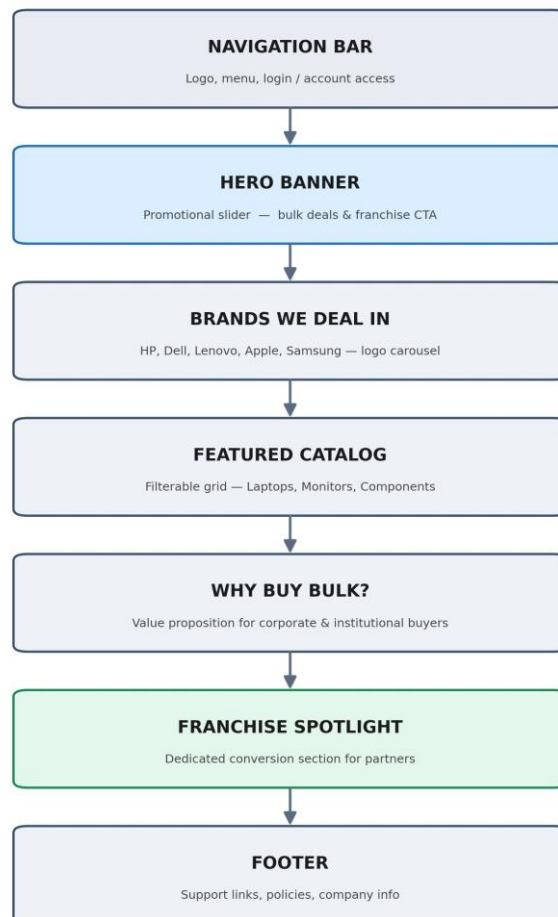


Figure 1 — End-to-end landing page section flow

3. Section-by-Section Breakdown

3.1 Hero Banner

Visual: Full-width, light-themed promotional banner with high-resolution IT hardware imagery and smooth slide transitions.

Messaging: “India’s Premier IT Hardware Bulk & Franchise Ecosystem.”

CTA 1 (Sky Blue): “Explore Bulk CatLog” — smooth-scrolls to the product catalog.

CTA 2 (Emerald Green): “Apply for Franchise” — smooth-scrolls to the Franchise Spotlight section.

3.2 Brands We Deal In

Layout: Auto-scrolling horizontal logo carousel of top IT manufacturers (HP, Dell, Lenovo, Apple, Samsung, etc.).

Interactivity: Hovering over a brand logo filters the catalog below to show only that manufacturer's hardware.

3.3 Bulk Product Catalog

Layout: Clean 4-column grid covering laptops, monitors, motherboards, RAM, and storage.

Normal / unregistered users: See standard retail pricing, or a "Login to View Bulk B2B Rates" label.

Franchise partners: See their exclusive, highly discounted factory rates in place of retail pricing.

3.4 Why Choose Us for Bulk Orders?

100% Genuine Hardware: Direct manufacturer billing with full brand warranty.

Pan-India Logistics: Secure, insured transit for high-value bulk IT shipments.

Dedicated Account Manager: Single point of contact for corporate procurement.

3.5 Franchise Portal Spotlight

Purpose: Engineered to convert local entrepreneurs into franchise owners by clearly outlining partnership value.

Real / Factory Pricing: Franchisees bypass distributor chains, buying at actual wholesale/import cost.

Territory Monopoly: No other official franchise opens within an approved city/zone.

Setup Support: Store layout guidelines plus digital branding assets.

3.6 Footer

Contents: Support links, company policies, and general company information.

5. Three-Portal Access Control Model

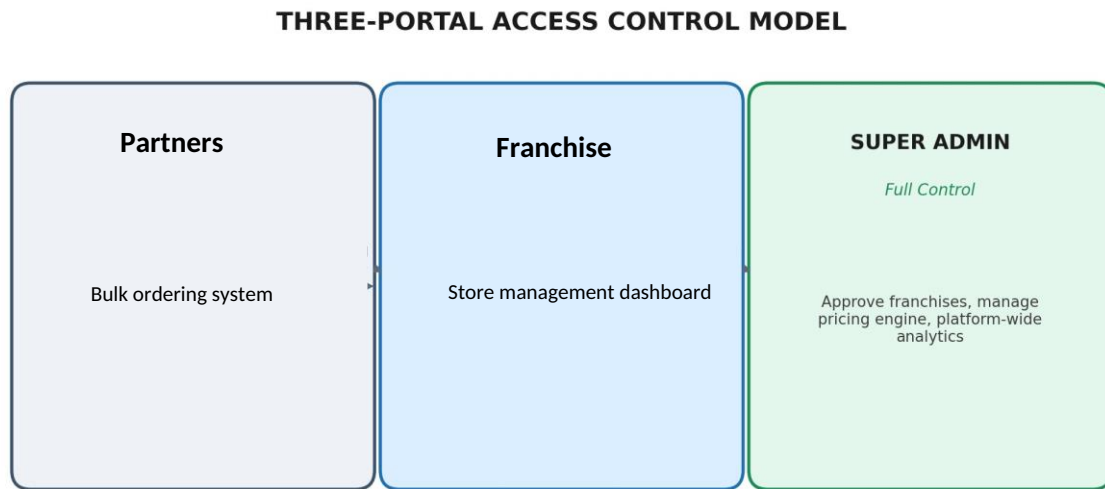


Figure 3 — Role hierarchy and escalation path

6.1 Partners (With / Without Login)

Without login: visitors can browse the site, read about the company, view hardware specifications, and read the Franchise Spotlight section.

With login: registered standard users can save wishlists, track standard retail orders, and submit a franchise application. Can order the items in bulk. Partners will provide RFP.

6.2 Franchise Partner Portal (The Store Owner Dashboard)

Once an applicant is approved by the admin, the account upgrades to Franchise Partner status and the interface transforms:

- **Franchise Growth Dashboard** — shows current credit limit, pending bulk shipments, warranty claim statuses, and monthly sales targets.

6.3 Admin Portal

Isolated from the public storefront, the admin dashboard provides absolute platform control:

- **Franchise Application Workflow** — review incoming applications, filter by city to avoid overlapping territories, and one-click approve to auto-issue login credentials and unlock real pricing.

- **Dynamic Price & Stock Engine** — set global discount percentages for franchise tiers or adjust real prices based on daily market fluctuations.
- **Platform Analytics** — view total bulk volume moved, top-performing city franchises, and low-stock alerts.

Technology Stack

The platform utilizes the PERN Stack to deliver a high-performance, secure, and scalable B2B e-commerce and franchise management ecosystem.

1. Frontend Layer (React.js & Tailwind CSS)

- **React.js**: Engineered as a Single Page Application (SPA) using modular components (sticky nav, dynamic product grids, brand carousels) for seamless state management across all three portal dashboards without page reloads.
- **Tailwind CSS**: Enables a modern, lightweight, and strictly light-themed UI. It powers smooth custom animations (hover elevations, slide-out cart drawers) while maintaining fast rendering speeds.

2. Backend Layer (Node.js & Express.js)

- **Node.js**: An asynchronous runtime ensuring zero performance bottlenecks during concurrent API requests, live search auto-completion, and real-time inventory checks.
- **Express.js**: Handles RESTful API routing and implements strict Role-Based Access Control (RBAC)—dynamically serving standard prices to normal users, unlocking discounted "Real/Factory Rates" for Franchise Partners, and isolating full control for the Super Admin.

3. Database Layer (**PostgreSQL**)

- **ACID Compliance**: Eliminates inventory discrepancies during high-volume B2B transactions.
- **Relational & JSONB Mapping**: Efficiently connects complex data (*Users* —► *Territories* —► *Orders*) while utilizing JSONB to natively store flexible, product-specific IT hardware specifications (CPU, RAM, Display specs) within SQL tables.